

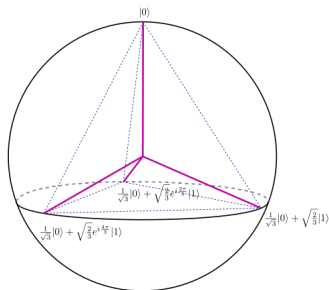
A ququart SIC for ion trap quantum computers?

Matthew B. Weiss

UMB: QBism Group

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Symmetric informationally complete sets



Fix a Hilbert space $\mathcal{H}_d$. If you can find a set of d^2 rank-1 projectors $\{\Pi_i\}$ satisfying

$$\text{tr}(\Pi_i \Pi_j) = \frac{d\delta_{ij} + 1}{d + 1}, \quad (1)$$

then you've found regular simplex inscribed in quantum state-space: a SIC [6]!

SIC's

- ▶ SIC states (with a group covariance) have *maximal magic* [5].
- ▶ The corresponding POVM $\{E_i = \frac{1}{d}\Pi_i\}$ is *optimal for linear quantum state tomography* [7].
- ▶ A SIC-based generalization of the double-split experiment demonstrates the *inherent contextuality* of quantum theory [8, 1].
- ▶ The numbers involved in constructing a SIC have *deep algebraic number theoretic significance* [4].
- ▶ The question of whether SIC's exist in all dimensions directly relates to Hilbert's 12 problem about abelian extensions of the rationals [3]!

Implementation

Is there a general method for implementing SIC-POVM's?

First, let's consider its symmetries.

The Weyl-Heisenberg Group

- ▶ Save one sporadic case, known SICs are “WH-covariant.”
- ▶ Fixing $\mathcal{H}_d$, let $\omega = e^{2\pi i/d}$ and define clock and shift operators,

$$Z|m\rangle = \omega^m|m\rangle \quad X|m\rangle = |m+1\rangle \quad X = F^\dagger ZF, \quad (2)$$

where F is the DFT, as well as Weyl-Heisenberg displacement operators,

$$D_{a,b} = X^a Z^b. \quad (3)$$

- ▶ The d^2 SIC states arise as the orbit of a “fiducial” state Π ,

$$\Pi_{a,b} = D_{a,b} \Pi D_{a,b}^\dagger. \quad (4)$$

WH group covariance means:

- ▶ We can find fiducials more easily classically.
- ▶ We can implement SICs more easily quantum mechanically.

Historical flavor

From *Bell System Technical Journal*:

B.S.T.J. BRIEFS

On the Simultaneous Measurement of a Pair of Conjugate Observables

By E. ARTHURS and J. L. KELLY, JR.

(Manuscript received December 16, 1964)

A precise theory of the simultaneous measurement of a pair of conjugate observables is necessary for obtaining the classical limit from the quantum theory, for determining the limitations of coherent quantum mechanical amplifiers, etc. The uncertainty principle, of course, does

Arthurs-Kelly procedure

- ▶ Prepare two ancillas in a special starting state.
- ▶ Coherently shift ancilla 1 conditional on the position of the system.
- ▶ Coherently shift ancilla 2 conditional on the momentum of the system.
- ▶ Measure the ancillas.

Realizes a “Weyl-Heisenberg covariant POVM” on the system.

- ▶ E.g., a coherent state measurement, or “heterodyne” measurement in optics.
- ▶ Can be adapted to qudits: in particular, for a SIC measurement.

Qudit Arthurs-Kelly

- ▶ Let $\{|k\rangle\}$ denote discrete position states (the computational basis).
- ▶ Let $\{|m\rangle_p = F|m\rangle\}$ denote discrete momentum states.
- ▶ The Arthurs-Kelly unitary is:

$$U_{AK} = \left(\sum_m I \otimes X^{-m} \otimes |m\rangle_p \langle m|_p \right) \left(\sum_k X^{-k} \otimes I \otimes |k\rangle \langle k| \right). \quad (5)$$

- ▶ But how to prepare the ancillas given the desired fiducial state?

Qudit Arthurs-Kelly

- ▶ To prepare the “ready state” of the two ancillas,

$$U_{PA} = \left(\sum_j |j\rangle\langle j| \otimes Z^j \otimes I \right) (I \otimes F^\dagger \otimes I). \quad (6)$$

- ▶ Let $|\phi\rangle$ be a SIC fiducial, and $|\psi\rangle$ the system state, then

$$|\text{post-interaction}\rangle = U_{AK} U_{PA} \left(|\phi^*\rangle \otimes |\phi\rangle \otimes |\psi\rangle \right). \quad (7)$$

- ▶ Measuring the two ancillas gives d^2 possible outcomes with

$$P(a, b) = \frac{1}{d} \text{tr}(\Pi_{a,b} \rho). \quad (8)$$

- ▶ Afterwards, the system is projected into the SIC state $\Pi_{a,b}$.

But actually, there's a simpler way.

Thrifty SIC

- ▶ Let $|\psi\rangle$ an arbitrary state, and $|\phi\rangle$ be the fiducial.

$$\begin{aligned} & \left(\langle a| \otimes \langle b| \right) (I \otimes F^\dagger) \left(\sum_j X^{-j} \otimes |j\rangle\langle j| \right) \left(|\psi\rangle \otimes |\phi^*\rangle \right) \\ &= \frac{1}{\sqrt{d}} \langle \phi | D_{a,b}^\dagger | \psi \rangle. \end{aligned} \tag{9}$$

- ▶ The probabilities for a computational basis measurement are precisely the SIC probabilities.
- ▶ Just one ancilla!
- ▶ But afterwards, the state is projected into a computational basis state.

Trade-offs

- ▶ The Arthurs-Kelly implementation requires two ancillas, and consequently a more involved interaction.
- ▶ The “thrifty” implementation requires just one ancilla.
- ▶ The downside is that the system is not projected into a SIC state afterwards: relevant when you are measuring a subsystem of a multipartite system.
- ▶ The upside is you can do cheaper tomography!

Why we care about a $d = 4$ SIC

A CONSTRUCTIVE APPROACH TO ZAUNER'S CONJECTURE VIA THE STARK CONJECTURES

MARCUS APPLEBY, STEVEN T. FLAMMIA, AND GENE S. KOPP

ABSTRACT. We propose a construction of d^2 complex equiangular lines in $\mathbb{C}^d$, also known as SICs or SIC-POVMs, which were conjectured by Zauner to exist for all d . The construction gives a putatively complete list of SICs with Weyl–Heisenberg symmetry in all dimensions $d > 3$. Specifically, we give an explicit expression for an object that we call a ghost SIC, which is constructed from the real multiplication values of a special function and which is Galois conjugate to a SIC. The special function, the Shintani–Faddeev modular cocycle, is more precisely a tuple of meromorphic functions indexed by a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$. We prove that our construction gives a valid SIC in every case assuming two conjectures: the order 1 abelian Stark conjecture for real quadratic fields and a special value identity for the Shintani–Faddeev modular cocycle. The former allows us to prove that the ghost and the SIC are Galois conjugate over an extension of $\mathbb{Q}(\sqrt{\Delta})$ where $\Delta = (d+1)(d-3)$, while the latter allows us to prove idempotency of the presumptive fiducial projector. We provide computational tests of our SIC construction by cross-validating it with known solutions, particularly the extensive work of Scott and Grassl, and by constructing four numerical examples of nonequivalent SICs in $d = 100$, three of which are new. We further consider rank- r generalizations called r -SICs given by maximal equichordal configurations of r -dimensional complex subspaces. We give similar conditional constructions for r -SICs for all r, d such that $r(d-r)$ divides $(d^2 - 1)$. Finally, we study the structure of the field extensions conjecturally generated by the r -SICs. If K is any real quadratic field, then either every abelian Galois extension of K , or else every abelian extension for which 2 is unramified, is generated by our construction; the former holds for a positive density of field discriminants.

Why we care about a $d = 4$ SIC

- ▶ $d = 4$ is the first dimension for which the amplitudes defining a SIC have a distinctive number theoretic character.
- ▶ The number theory provides a systematic framework for constructing SICs: a tantalizing intersection of pure math and physics whose ramifications have yet to be fully explored.
- ▶ Also, of course, $d = 4$ is the first dimension you can compare two qubit tomography to ququart tomography. Qubits are not the future!
- ▶ Finally, we want to do SICs not because they are easy, but because they are hard! To stretch the technology, provide a stress test for quantum hardware.

A $d = 4$ fiducial

Exploit the “monomial representation” [2] to write

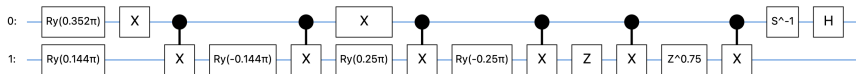
$$|\phi\rangle =$$

$$(H \otimes I) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{i\pi(-1/4)} & 0 & 0 \\ 0 & 0 & e^{i\pi(1/4)} & 0 \\ 0 & 0 & 0 & e^{i\pi(1/2)} \end{pmatrix} \frac{1}{\sqrt{5 + \sqrt{5}}} \begin{pmatrix} \sqrt{2 + \sqrt{5}} \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad (10)$$

where H is the qubit Hadamard gate.

A $d = 4$ fiducial

As a circuit on two qubits:



Ion traps

- ▶ In “Experimental single-setting quantum state tomography”, you implement a qubit SIC using a native quqart *locally*.
- ▶ Can we similarly realize a ququart SIC using a sixteen level ion? Or with two (or three) native ququarts?
- ▶ In “A universal qudit quantum processor with trapped ions,” you implement $d = 7$ qudits, and show in general how to implement a controlled exchange gate C_{EX} from which one can implement C_{SUM} , that is, the controlled shift CX .
- ▶ If doing CX , CZ , and F are easy, then the only difficulty is in preparing the fiducial itself!

Take 3: A block-diagonal scheme

- ▶ Both of the methods presented so far separate out the preparation of the fiducial from the implementation of the measurement itself.
- ▶ Exploiting WH symmetry, one may more directly construct a unitary which implements a SIC: even better, the unitary can be block-diagonalized in a straightforward way.
- ▶ In particular, the method can be tailored to the monomial representation of the $d = 4$ fiducial.
- ▶ (Originally developed for a quantum optics set-up.)

Take 3: A block-diagonal scheme

Let $\lambda = e^{i\pi/4}$, and define gates

$$D = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, L_{\pm} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mp \lambda & 1 \\ \pm \lambda & 1 \end{pmatrix}$$

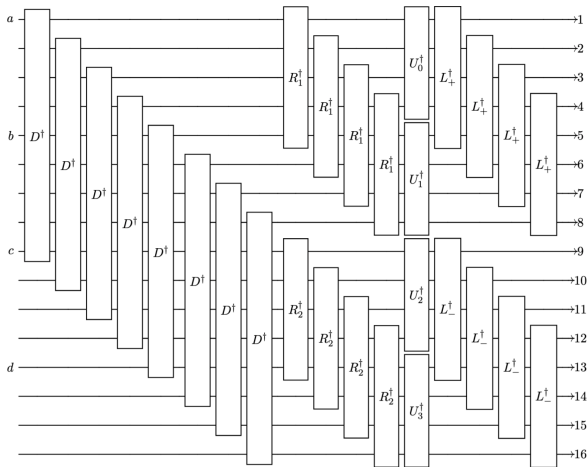
$$R_1 = \frac{1}{\sqrt{2}\lambda} \begin{pmatrix} 1 & -1 \\ \lambda & \lambda \end{pmatrix}, R_2 = \frac{1}{\sqrt{2}\lambda} \begin{pmatrix} \lambda & \lambda \\ 1 & -1 \end{pmatrix},$$

$$U_{0/1} = \frac{1}{\sqrt{2}} \begin{pmatrix} u \pm i\lambda & -1 \pm i\lambda & -iu \mp \lambda & i \mp \lambda \\ -1 \mp i\lambda & -u \pm i\lambda & i \pm \lambda & iu \mp \lambda \\ -1 \pm i\lambda u & -1 \mp i\lambda & i \mp \lambda u & i \pm \lambda \\ -1 \pm i\lambda & 1 \pm i\lambda u & i \mp \lambda & -i \mp \lambda u \end{pmatrix}$$

$$U_{2/3} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \pm \lambda & i(1 \pm \lambda u) & i(1 \mp \lambda) & -1 \pm \lambda u \\ 1 \pm \lambda u & -i(1 \pm \lambda) & i(1 \mp \lambda u) & 1 \mp \lambda \\ -1 \pm \lambda & i(u \mp \lambda) & -i(1 \pm \lambda) & -u \mp \lambda \\ u \mp \lambda & i(1 \mp \lambda) & i(u \pm \lambda) & -1 \mp \lambda \end{pmatrix}$$

Take 3: A block-diagonal scheme

Suppose the ion has 16 energy levels, and the state to be measured is prepared as a superposition of every 4th level (a, b, c, d).



Here U_0, U_1, U_2, U_3 act on four degrees of freedom. The other gates act only the first and last degrees of freedom encompassed by the gate.

Now we know how to perform SIC's.

But what can we do with them?

From tomography to the Born rule

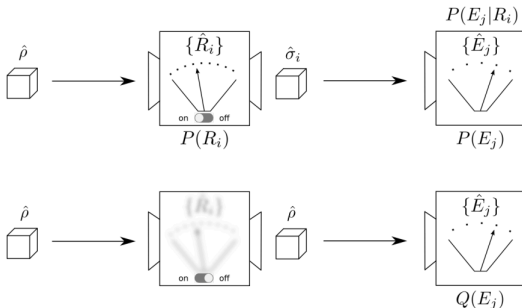
- Reconstructing density matrices from SIC probabilities:

$$\rho = \sum_i \left[(d+1)P(R_i) - \frac{1}{d} \right] \Pi_i, \quad (11)$$

where $P(R_i) = \frac{1}{d} \text{tr}(\Pi_i \rho)$.

- Leads to an elegant reformulation of the Born rule as a consistency rule for probability assignments.
- Checking that probability assignments are consistent according to this rule provides a way to validate quantum hardware!

The Born Rule



$$P(E_j) = \sum_i P(E_j|R_i)P(R_i) \quad (12)$$

$$Q(E_j) = \sum_{ik} P(E_j|R_i)\Phi_{ik}P(R_k) \quad (13)$$

where $\Phi = P(R|R)^{-1}$ for any linearly independent $\{R_i\}$ and $\{\sigma_i\}$.

The Born Rule

- ▶ In particular, for a SIC:

$$Q(E_j) = \sum_i P(E_j|R_i) \left[(d+1)P(R_i) - \frac{1}{d} \right]. \quad (14)$$

- ▶ Among all IC-POVMs with d^2 elements, $\|I - \Phi\|$ achieves its minimum only for SIC's.
- ▶ Other measurements artificially exaggerate the difference between classical and quantum! In this way, the SICs provide a good standard measurement.
- ▶ Let $W(E_j) = (d+1)P(R_i) - \frac{1}{d}$: these are quasiprobabilities.
- ▶ Their negativity diagnoses contextuality.
- ▶ The more noise, the less negativity.

Consistency

Let us consider the following concrete set of experiments:

1. Prepare the SIC states in turn, and then a SIC measurement.
Yields matrix of probabilities $P_{ij} = P(R_i|R_j)$.
2. Prepare the computational basis states $\{\Pi_i = |i\rangle\langle i|\}$, and then performing a SIC. Yields probabilities $p_{ij} = P(R_j|\Pi_i)$.
3. Prepare the SIC states, and a computational basis measurement: $C_{ij} = P(\Pi_i|R_j)$.
4. Prepare the computational basis states, and then a computational basis measurement: $q_{ij} = P(\Pi_i|\Pi_j)$.

Letting $\Phi = P^{-1}$, the Born rule then demands the following consistency criterion,

$$q = C\Phi p. \tag{15}$$

How well can we approximate this in the presence of noise? To what extent can we minimize $\|q - C\Phi p\|$?

Metrics of success

- ▶ $\|P_{\text{SIC}} - P\|$
- ▶ $\|\Phi_{\text{SIC}} - \Phi\|$
- ▶ $\|I - \Phi\|$
- ▶ $\|q - C\Phi p\|$
- ▶ Average negativity of quasiprobabilities over random states.
- ▶ Further, compare performance against other WH-POVM's with less magic (that is, for alternative choices of fiducials).
- ▶ If enough precision is available, could we use variational algorithms to *learn* SIC fiducials themselves by tuning ansatz parameters?

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